

EE460 Spacecraft Project Document

Spring 2024

1.0 Objective

The goal of this project is to construct a control law in Simulink® which will maneuver a simulated spacecraft to 4 different points in a limited amount of time with a limited amount of fuel. The purpose of visiting each point is to align an antenna to send/receive information. To accomplish this mission, the spacecraft needs to maintain its position at each point within specified tolerances for a specified time period.

2.0 Spacecraft Model

A model of the spacecraft has been created and is provided to the students. Note that this model is a fictional creation that does not represent any particular spacecraft. Also note that while some effort was taken to use realistic numbers, not all values are realistic. This was done to facilitate the goals for this project. These goals included a simulation which could be completed in two minutes of real time using a display that is scaled to be readable to the user and has mass properties which vary a non-trivial amount.

The spacecraft is modeled as a three degree of freedom system constrained to move in a plane. The three degrees of freedom are: translation in the X-direction, translation in the Y-direction, and rotation about the Z-axis.

2.1 State-Equations

A state-space model of the plant can be created with six states and three pseudo-control input values.

The state vector, $\bar{x}$:

$$\bar{x} = \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} x \text{ position} \\ x \text{ velocity} \\ y \text{ position} \\ y \text{ velocity} \\ \text{angular position} \\ \text{angular rate} \end{bmatrix}$$

The input vector of pseudo-controls, $\bar{u}_p$:

$$\bar{u}_p = \begin{bmatrix} F_x \\ F_y \\ M_z \end{bmatrix} = \begin{bmatrix} \text{Force in the inertial } X - \text{axis} \\ \text{Force in the inertial } Y - \text{axis} \\ \text{Moment about the inertial } Z - \text{axis} \end{bmatrix}$$

For fixed mass properties, the state equations are:

$$\dot{\bar{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \bar{x} + \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{M} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{M} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{J} \end{bmatrix} \bar{u}_p$$

In actuality, the spacecraft does not have the three controls listed. Instead, it has eight thrusters, each of which is capable of generating a modulated amount of thrust between zero and one hundred Newtons. It is easier to model the forces due to the thrusters in the body axis instead of the inertial axis system. The relationship between the pseudo-controls, $\bar{u}_p$, and the body axis forces and moment is shown below.

$$\bar{u}_p = \begin{bmatrix} F_x \\ F_y \\ M_z \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} F_{x_B} \\ F_{y_B} \\ M_{z_B} \end{bmatrix}$$

The relationship between the thrust values, u_i , and the body axis forces and moment is shown in the equation below. Note that this uses nominal values for the center of gravity location, the centroid of the spacecraft, and the moment arms.

$$\begin{bmatrix} F_{x_B} \\ F_{y_B} \\ M_{z_B} \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 1 & -1 \\ -1 & 1 & -1 & 1 & 3 & -3 & 3 & -3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \end{bmatrix}$$

Figure 1 below shows how the eight thrusters and the four fuel tanks are configured.

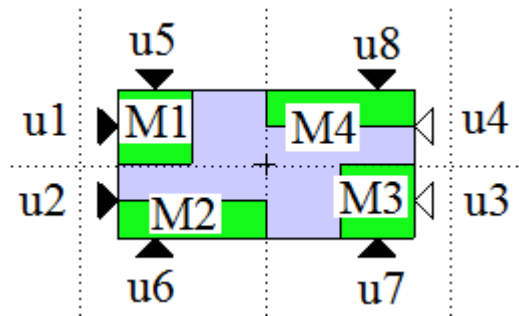


Figure 1: Thrusters and Fuel Tank Numbering

The centroid of the spacecraft is marked with a + sign. The orientation of the vehicle, θ , is measured between the X-inertial axis and the X-body axis. The X-body axis goes from the centroid to a point midway between thrusters three and four.

For constant mass properties, the relationship between these thrust values, u_i , and the body axis accelerations is shown in the equation below.

$$\begin{bmatrix} \ddot{x}_B \\ \ddot{y}_B \\ \ddot{\theta}_B \end{bmatrix} = \begin{bmatrix} \frac{1}{M} & \frac{1}{M} & -\frac{1}{M} & -\frac{1}{M} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{M} & \frac{1}{M} & \frac{1}{M} & -\frac{1}{M} \\ -\frac{1}{J} & \frac{1}{J} & -\frac{1}{J} & \frac{1}{J} & \frac{1}{J} & -\frac{1}{J} & \frac{1}{J} & -\frac{1}{J} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \end{bmatrix}$$

One method for allocating the controls is to find the desired body axis accelerations and use a pseudo-inverse on the matrix shown above. A pseudo-inverse is a poor solution for this problem because each of the control effectors is limited to only positive values. Several methods for dealing with this complexity will be discussed in class.

2.2 Mass Properties

The spacecraft contains four fuel tanks which each hold a maximum of 10kg of fuel.

$$M = M_{base} + M_1 + M_2 + M_3 + M_4$$

$$M_{base} = 60 \text{ kg}$$

$$M_i = \text{mass in fuel tank } i$$

$$0 \leq M_i \leq 10 \text{ kg}$$

The equations below are used to calculate the center of mass and mass moment of inertia for the vehicle as a function of fuel load. First, the mass moment of inertia is calculated about the centroid, then the parallel axis theorem is used to find it for the present center of gravity location.

$$J = J_{base} + J_{fuel} + Md^2$$

$$J_{base} = 140 \text{ kg} \cdot \text{m}^2$$

$$J_{fuel} = \frac{17M_1 + 8M_2 + 17M_3 + 8M_4}{12} + 10M_1 + 6.25M_2 + 10M_3 + 6.25M_4$$

$$X_{CG} = \frac{-3M_1 - 2M_2 + 3M_3 + 2M_4}{M}$$

$$Y_{CG} = \frac{M_1 - 1.5M_2 - M_3 + 1.5M_4}{M}$$

$$d^2 = X_{CG}^2 + Y_{CG}^2$$

X_{CG} and Y_{CG} are in the body axis system which is centered at the centroid of the vehicle. The mass properties of the plant will vary as fuel is burned. One of the design choices for the control law regards the modeling of the mass properties within the controller. One option is to use constant mass properties in the control law. Another option is to have mass properties which vary as fuel is burned. If you are using constant mass properties, select a set of fuel characteristics and find the associated mass properties. You do not need to select the full fuel condition, (i.e. you may find using the 50% or 73.2% fuel condition works better for more cases). If you have varying mass properties, you do not need to model all the mass properties changing. For example you may choose to only model the total mass change, and assume constant center of gravity and constant mass moment of inertia.

For reference, the Table 1 below shows the mass properties for several fuel loadings. Note that this table assumes fuel is burned equally from all tanks, which will result in no center of gravity changes.

Fuel State (%)	Mass (kg)	J (kg m ²)
100	100	506.67
75	90	415
50	80	323.33
25	70	231.67
0	60	140

Table 1: Mass Properties

2.3 Thrust Modeling

Each of the eight thrusters is capable of modulating the thrust produced between values of 0 to 100 N. The rate at which mass leaves the fuel tank is proportional to the thrust produced.

$$\dot{m}_j = \frac{u_i}{300}$$

Where u_i is thrust in N and $\dot{m}_j$ is the mass flow rate in kg/s.

Thruster failures are modeled in the plant. A thruster may fail in one of three possible ways: fail fully closed ($u_i = 0$), fail fully open ($u_i = 100$), or fail frozen at current value. If the thruster value differs from the commanded value for a sufficiently long period of time, the Health Monitoring System (HMS) will detect a failure and report that information to the control law via a logical variable. If a fuel tank becomes empty, the thrusters connected to it will no longer be capable of producing thrust. If this happens, the HMS will report that both thrusters connected to the empty fuel tank have failed and their value is zero. See Section 6 for more information on thruster data communicated to the control law.

2.4 Sensor Modeling

The spacecraft is equipped with nine different sensors. There are three possible design choices regarding sensor configurations. Each student will be assigned one of these configurations and will set the variable Sconfig to the appropriate value for their configuration.

The first sensor configuration, Sconfig=1, contains one position sensor, one velocity sensor, and one acceleration sensor for each axis. The second sensor configuration, Sconfig=2, contains two position sensors, and one velocity sensor for each axis. The third sensor configuration, Sconfig=3, contains three position sensors for each axis.

Sensor failures are modeled in the plant. A sensor may fail in one of three possible ways: fail to the minimum of its range, fail to the maximum of its range, fail frozen at current value. If the sensor value does not change for a sufficiently long period of time, the Health Monitoring System (HMS) will detect a failure and report that information to the control law via a logical variable. See Section 6 for more information on sensor data communicated to the control law.

2.5 Discrete Time Frame Rates

The plant is modeled as a discrete time system with a time step of 0.001 seconds. The variable dt0 is reserved for this time step. The control laws which are to be developed are to run with a time step no smaller than 0.01 seconds. The variable dt is reserved for this time step. Step sizes within the control law must be integer multiples of dt.

3.0 Control Law Command Strategies

Each point that the spacecraft needs to visit is defined by three coordinates, $P_i = [X \ Y \ \theta]$, and tolerances of these values, $P_{i_{tol}} = [\pm X \ \pm Y \ \pm \theta]$. The X and Y tolerances describe an ellipse (not a rectangle). For example, consider Point 1 defined by:

$$P_1 = [X \ Y \ \theta] = [0 \ 15 \ 45]$$
$$P_{1_{tol}} = [\pm X \ \pm Y \ \pm \theta] = [\pm 1 \ \pm 0.5 \ \pm 8]$$

As long as the centroid is within the ellipse, and the angle is within tolerance, communication is possible. See Figure 2 below. Note that the four points are color coded:

- P1 is red
- P2 is yellow
- P3 is green
- P4 is cyan

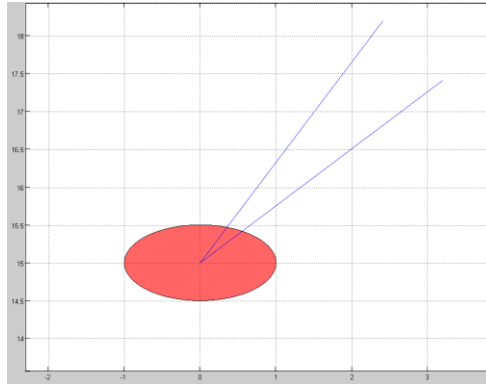


Figure 2: Representation of Point with Tolerances

There are different options for command strategies: Step Command, Ramp Command, Cubic Spline, Bang-Bang Control, and Dead-Band Control. Each of these will be discussed in class lectures and students will implement several of these for one of the homework assignments. However, the final project control law, with some exceptions, should use only one of the strategies.

4.0 User Interfaces

A requirement of this project is for each student to design a user interface which allows for a human operator to select the order of the four points to be visited.

For example:

1. Human operator uses a joystick with buttons to select each point.
2. Human operator uses a GUI to select each point.

If students have ideas on developing other user interfaces, see the course instructor to discuss them. It is recommended that the user interface include a method for adjusting the command values to account for possible sensor bias errors. Several interface options will be discussed in class lectures, however students are encouraged to investigate other options.

X.X Cheat Codes

Some students have asked if they could run an optimization code to select the order of the points. Since part of the project is to evaluate how an operator interface affects the overall performance of the controller, it is not permissible to by-pass the requirement of writing a user interface. However, it is permissible, if a student wishes, to add to the interface a series of commands (which do not interfere with the normal operation of the interface) that will run a special section of code. This specialized section of code may do things like find an optimal path through the points, or change the command types. For example, a student may choose to implement code which will engage/disengage a manual control feature.

5.0 Score

The score for a mission is calculated using the formula below:

$$\begin{aligned} \text{Score} = & 4(\text{Seconds of Time Remaining}) + 10(\text{Kilograms of Fuel Remaining}) \\ & + 100(\# \text{ Transmissions Completed}) \\ & - 100(\# \text{ Transmissions Not Complete}) \end{aligned}$$

6.0 Interface Control Documents

Data transmitted from the control law to the plant is communicated on a bus named UCMD_BUS. The contents of the bus must match the specifications in the Table 2 below. The current design does not use U9cmd or U10cmd, but the data bus includes them because the interface was designed to allow room for growth or last minute design changes. The data on this bus is sent 100 times per second.

Variable Name	Units	Range	Data Type	Scale Factor	LSB	Description
U1cmd	Newtons	[0 100]	UINT8	255/100	1/2.55 N	Thruster 1 command
U2cmd	Newtons	[0 100]	UINT8	255/100	1/2.55 N	Thruster 2 command
U3cmd	Newtons	[0 100]	UINT8	255/100	1/2.55 N	Thruster 3 command
U4cmd	Newtons	[0 100]	UINT8	255/100	1/2.55 N	Thruster 4 command
U5cmd	Newtons	[0 100]	UINT8	255/100	1/2.55 N	Thruster 5 command
U6cmd	Newtons	[0 100]	UINT8	255/100	1/2.55 N	Thruster 6 command
U7cmd	Newtons	[0 100]	UINT8	255/100	1/2.55 N	Thruster 7 command
U8cmd	Newtons	[0 100]	UINT8	255/100	1/2.55 N	Thruster 8 command
U9cmd	TBD	TBD	UINT8	255/100	TBD	Unused
U10cmd	TBD	TBD	UINT8	255/100	TBD	Unused

Table 2: Data send from the Control Law to the Plant

Data transmitted from the plant to the control law is communicated on a bus named Y_BUS. The contents of the bus match the specifications in the tables below. This data is transmitted to the control law 100 times per second.

There are three different sensor configurations. The first nine signals in Y_BUS correspond to these sensor values. For each sensor configuration, sensors one to three correspond to the X-axis sensors, sensors four to six correspond to the Y-axis sensors, and sensors seven to nine correspond to the rotation about the Z-axis.

The first sensor configuration, Sconfig=1, contains one position sensor, one velocity sensor, and one acceleration sensor for each axis. See Table 3 below. The second sensor configuration, Sconfig=2, contains two position sensors, and one velocity sensor for each axis. See Table 4 below. The third sensor configuration, Sconfig=3, contains three position sensors for each axis. See Table 5 below.

Variable Name	Units	Range	Data Type	Scale Factor	LSB	Description
Sensor1	Meters	[-100 100]	INT16	200	0.005	X Position
Sensor2	Meters/second	[-100 100]	INT16	100	0.01	X Velocity
Sensor3	Meters/second ²	[-4 4]	INT16	250	0.004	X Acceleration
Sensor4	Meters	[-100 100]	INT16	200	0.005	Y Position
Sensor5	Meters/second	[-100 100]	INT16	100	0.01	Y Velocity
Sensor6	Meters/second ²	[-4 4]	INT16	250	0.004	Y Acceleration
Sensor7	Degrees	[-600 600]	INT16	50	0.02	θ Position
Sensor8	Degrees/second	[-200 200]	INT16	100	0.01	θ Velocity
Sensor9	Degrees/second ²	[-360 360]	INT16	2.5	0.4	θ Acceleration

Table 3: Sensor configuration 1 Data sent from the Plant to the Control Law

Variable Name	Units	Range	Data Type	Scale Factor	LSB	Description
Sensor1	Meters	[-100 100]	INT16	200	0.005	X Position
Sensor2	Meters	[-100 100]	INT16	200	0.005	X Position
Sensor3	Meters/second	[-100 100]	INT16	100	0.01	X Velocity
Sensor4	Meters	[-100 100]	INT16	200	0.005	Y Position
Sensor5	Meters	[-100 100]	INT16	200	0.005	Y Position
Sensor6	Meters/second	[-100 100]	INT16	100	0.01	Y Velocity
Sensor7	Degrees	[-600 600]	INT16	50	0.02	θ Position
Sensor8	Degrees	[-600 600]	INT16	50	0.02	θ Position
Sensor9	Degrees/second	[-200 200]	INT16	100	0.01	θ Velocity

Table 4: Sensor configuration 2 Data sent from the Plant to the Control Law

Variable Name	Units	Range	Data Type	Scale Factor	LSB	Description
Sensor1	Meters	[-100 100]	INT16	200	0.005	X Position
Sensor2	Meters	[-100 100]	INT16	200	0.005	X Position
Sensor3	Meters	[-100 100]	INT16	200	0.005	X Position
Sensor4	Meters	[-100 100]	INT16	200	0.005	Y Position
Sensor5	Meters	[-100 100]	INT16	200	0.005	Y Position
Sensor6	Meters	[-100 100]	INT16	200	0.005	Y Position
Sensor7	Degrees	[-600 600]	INT16	50	0.02	θ Position
Sensor8	Degrees	[-600 600]	INT16	50	0.02	θ Position
Sensor9	Degrees	[-600 600]	INT16	50	0.02	θ Position

Table 5: Sensor configuration 3 Data sent from the Plant to the Control Law

The control law receives data from a Health Monitoring System which attempts to identify if failures have occurred. This data includes logical variables which are true when a failure has been detected as well as estimated values for failed thruster controls. See Table 6 below.

Variable Name	Units	Range	Data Type	Scale Factor	LSB	Description
S1fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
S2fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
S3fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
S4fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
S5fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
S6fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
S7fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
S8fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
S9fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U1fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U2fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U3fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U4fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U5fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U6fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U7fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U8fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U9fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U10fail	N/A	[0 1]	Boolean	N/A	1	True if failure detected
U1est	Newtons	[0 100]	UINT8	1	1	Estimated Thrust during failure
U2est	Newtons	[0 100]	UINT8	1	1	Estimated Thrust during failure
U3est	Newtons	[0 100]	UINT8	1	1	Estimated Thrust during failure
U4est	Newtons	[0 100]	UINT8	1	1	Estimated Thrust during failure
U5est	Newtons	[0 100]	UINT8	1	1	Estimated Thrust during failure
U6est	Newtons	[0 100]	UINT8	1	1	Estimated Thrust during failure
U7est	Newtons	[0 100]	UINT8	1	1	Estimated Thrust during failure
U8est	Newtons	[0 100]	UINT8	1	1	Estimated Thrust during failure
U9est	TBD	TBD	UINT8	1	TBD	Unused
U10est	TBD	TBD	UINT8	1	TBD	Unused

Table 6: Data from the Health Monitoring System sent from the Plant to the Control Law

The Communication Subsystem (COMM) also sends information to the control law. Anytime the spacecraft is aligned within tolerances at one of the specified locations, the communication subsystem will attempt to establish a connection. If the entire transmission is completed, the COMM system will send a transmission successfully completed signal. If the transmission is interrupted prior to completion (i.e. the spacecraft moves outside the tolerances before the transmission can be completed) the COMM system must reinitialize the process and start over again. See Table 7 below.

Variable Name	Units	Range	Data Type	Scale Factor	LSB	Description
P1OK	N/A	[0 1]	Boolean	N/A	1	True if connection established
P1Done	N/A	[0 1]	Boolean	N/A	1	True if transmission completed
P2OK	N/A	[0 1]	Boolean	N/A	1	True if connection established
P2Done	N/A	[0 1]	Boolean	N/A	1	True if transmission completed
P3OK	N/A	[0 1]	Boolean	N/A	1	True if connection established
P3Done	N/A	[0 1]	Boolean	N/A	1	True if transmission completed
P4OK	N/A	[0 1]	Boolean	N/A	1	True if connection established
P4Done	N/A	[0 1]	Boolean	N/A	1	True if transmission completed

Table 7: COMM Data sent from the Plant to the Control Law

7.0 Files Available on Course Website

EE460_S3D_Plant.slx

This file contains a model of the plant. Students are not permitted to modify the plant model. When students submit their work, the plant will be replaced with the version contained within this file. Students should begin with this file, rename it LastName_FirstName_EE460S3D_R00.slx, and then add their control law to it. This will preserve some settings (for example integration type and step size) and will prevent some errors and warnings. The final two numbers in the name should be used as a revision number to keep track of your modifications.

INIT_S3D_01.m

This file contains variables which are used to initialize the plant model. The variable names in this file are reserved for these values. Students may use these variables in their control law, but do not assign different values to these variables. This file will be run during the initialization phase AFTER the student created initialization file.

GetLVL_S3D.m

This script file contains data for different levels. Initially, a file is provided which contains the four levels to be used for the homework assignments and a couple extras. Students are encouraged to add to this file to create their own levels to test out their control laws. For Homework 8 students will be required to submit a level that they created to test their system. Later, a revised version, GetLVL_S3D_ALL.m, will be made available which includes the levels created by the students in the class.

PlotS3D.m

This script file is used to present a graphical representation of the spacecraft and the communication points.

SigNameProp.m

This script file can be used to turn on or off propagated signal names for the entire block diagram. It is usefully for making sure your diagram adheres to the rule that all signals going into or out of subsystems are named.

Note that these files were saved as MATLAB R2023a. All files submitted by students must be compatible with MATLAB R2023a. This is the version of MATLAB that is available on the classroom and lab computers. If you do your development work in a different version, you are responsible for checking to make sure that it runs successfully on the specified version prior to submission.

Homework 5

Create a control law to work with the plant using the ideas already developed in the class. Using the third sensor configuration, Sconfig=3 (triplex redundant position sensors), use an observer to compute the estimated states, $\hat{x}$.

For this assignment you will use the Level 1 data. Fuel burn is turned off and you can assume constant mass properties for a fully fueled vehicle:

$$M = 100 \text{ kg}$$

$$J = 506 \frac{2}{3} \text{ kg} \cdot \text{m}^2$$

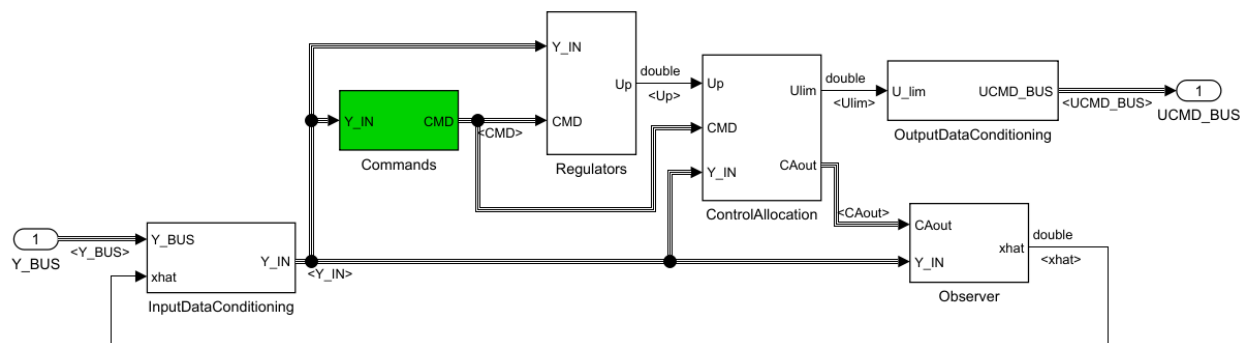
Note that the simulation does compute how much fuel would have been burned and records this information in the variable FuelBurned (kg) which is used to calculate your score.

The COMM points are defined by:

$$\begin{aligned} X_{cmd} &= [10 \quad -10 \quad 20 \quad -20] \\ Y_{cmd} &= [0 \quad 0 \quad 0 \quad 0] \\ \theta_{cmd} &= [0 \quad 0 \quad 0 \quad 0] \end{aligned}$$

See the file GetLVL_S3D.m for the complete set of data.

Modify Homework 4 by adding a Commands block to your control law. For each point, use a simple step command for $\bar{u}_r$ and go to the points in the order specified: P1 then P2 then P3 then P4. Use the signals from the plant to determine when a point is completed and then change the commands to go to the next point.



Example of Top Level Control Law Architecture

For Homework 5, submit a list of your preferences for:

user interfaces (ranked 1-3, 1 being most preferred)

variable or constant mass properties in the control law (ranked 1-2, 1 being most preferred)

User Interfaces (only rank one joystick option)

_____ Instructor provided Joystick

_____ Student provided Joystick or Game Controller

_____ GUI

_____ Other: Specify _____

Control Law Mass Properties

_____ Constant

_____ Variable

Submit via email by 11:59 PM on Friday March 22:

One Simulink file containing you simulation (modified from the provided plant model and named correctly)

One M-file which will initialize any variables used by your controller named:

LastName_FirstName_EE460_H5.m

One document or hard copy listing your preferences for User Interfaces, and Mass Properties.

Grading Rubric: 50 points total

Complete transmission at each of 4 points: 10 pts/each if completed, 5 pts if established communication but did not complete transmission. If submitted files do not run due to errors, 0/40 points will be awarded for this part of the assignment.

Block Diagram follows rules and paperwork submitted correctly: 10 pts

Homework 6

Modify the Homework 5 control law to go to the first point, P1, using a step command then go to the second point, P2, using a ramp command, go to P3 using a cubic spline, and go to P4 using a bang-bang control.

For this homework assignment, the course to be followed is defined by the Level 2 data:

$$\begin{aligned}X_{cmd} &= [0 \quad 0 \quad 20 \quad -20] \\Y_{cmd} &= [0 \quad 10 \quad 10 \quad 10] \\\theta_{cmd} &= [90 \quad 90 \quad 90 \quad 90]\end{aligned}$$

See the file GetLVL_S3D.m for the complete set of data.

For this homework assignment, you will change sensor configuration to use Sconfig=1. This is the case with a single position, velocity, and acceleration sensor for each axis. For this and all subsequent levels noise has been added to the sensor data. For this assignment you may use observer or sensed values. You may wish to modify your observer and/or filter the sensed values in an attempt to alleviate issues which arise from the sensor noise.

For Homework 6, list your preferences for:

sensor/observer configuration (ranked 1-6, 1 being most preferred)

command sequencing (ranked 1-2, 1 being most preferred)

command strategy (ranked 1-5, 1 being most preferred)

regulator architecture (ranked 1-4, 1 being most preferred)

Sensor/Observer Configurations:

- ☐ Use sensed values with Sconfig = 1
- ☐ Use sensed values with Sconfig = 2
- ☐ Implement a reduced order observer with Sconfig=3
- ☐ Implement a full order observer with Sconfig = 1
- ☐ Implement a full order observer with Sconfig = 2
- ☐ Implement a full order observer with Sconfig = 3

Command Sequencing:

- ☐ Combined Commands (a.k.a. all axes at once)
- ☐ Sequential Commands (translation and rotation done independently)

Command Strategies:

- _____ Step Command
- _____ Ramp Command
- _____ Cubic Command
- _____ Bang-Bang Commands
- _____ Dead-Band Commands

Regulator Architecture:

- _____ Proportional+Derivative (PD)
- _____ Proportional+Integral+Derivative (PID)
- _____ Proportional+Integral+Derivative+Feedforward (PID+F)
- _____ Proportional+Integral+Derivative+F(s) (PID+F(s))

Submit via email by 11:59 PM on Friday March 29:

One Simulink file containing you simulation (modified from the provided plant model and named correctly)

One M-file which will initialize any variables used by your controller named:

LastName_FirstName_EE460_H6.m

One document or hard copy listing your preferences for Sensor/Observer Configuration Command Sequencing, Command Strategy, and Regulator Architecture.

Grading Rubric: 50 points total

Complete transmission at each of 4 points: 10 pts/each if completed with the specified command strategy, 5 pts if established communication but did not complete transmission or if the command strategy was not properly implemented. If submitted files do not run due to errors, 0/40 points will be awarded for this part of the assignment.

Block Diagram follows rules and paperwork submitted correctly: 10 pts

Homework 7

Modify the Homework 6 control law to incorporate your user interface and all the options you were assigned from Homework 5 and Homework 6.

For this homework assignment, you will begin to use the sensor configuration which has been selected for your project. This scenario involves a sensor failure. You should begin to modify the control law to accommodate sensor failures.

Also, mass will vary as fuel is burned. If your control law design will incorporate a variable mass model, you should begin to incorporate that at this time.

For this homework assignment, the course to be followed is defined by the Level 3 data:

$$\begin{aligned}X_{cmd} &= [0 \quad 20 \quad 0 \quad -20] \\Y_{cmd} &= [15 \quad 0 \quad -15 \quad 0] \\\theta_{cmd} &= [45 \quad 180 \quad 135 \quad -90]\end{aligned}$$

See the file GetLVL_S3D.m for the complete set of data.

Submit via email by 11:59 PM on Friday April 5:

One Simulink file containing your simulation (modified from the provided plant model and named correctly)

One M-file which will initialize any variables used by your controller named:

LastName_FirstName_EE460_H7.m

One document or hard copy listing the day your preferences for the day to do your project presentation.

April 17, 19, 22, 24 (ranked 1-4, 1 being most preferred)

Grading Rubric: 50 points total

Complete transmission at each of 4 points: 10 pts/each if completed, 5 pts if established communication but did not complete transmission. If submitted files do not run due to errors, 0/40 points will be awarded for this part of the assignment.

Block Diagram follows rules and paperwork submitted correctly: 10 pts

If your submitted work does not incorporate the options you were assigned, 5 pts will be deducted for each option not used.

Homework 8

Modify the Homework 7 control law to accommodate thruster failures. Level 4 includes a thruster failure. For this homework assignment, you will continue to use the sensor configuration which has been selected for your project. The Level 4 scenario involves a sensor failure. Mass will vary as fuel is burned.

For this homework assignment, the course to be followed is defined by the Level 4 data:

$$\begin{aligned}X_{cmd} &= [-10 \quad 10 \quad 20 \quad -20] \\Y_{cmd} &= [-10 \quad -10 \quad -10 \quad 15] \\\theta_{cmd} &= [-180 \quad -45 \quad 90 \quad 45]\end{aligned}$$

See the file GetLVL_S3D.m for the complete set of data.

Design a Level to test your control laws according to the following rules:

$-20 \leq X_{cmd} \leq 20$: values can range from -20m to 20m.

$-15 \leq Y_{cmd} \leq 15$: values can range from -15m to 15m.

$-180^\circ \leq \theta_{cmd} \leq 180^\circ$: values can range from -180° to 180° .

The values for tolerances on X and Y must not be less than 0.2m and not greater than 2m.

The values for tolerances on θ must not be less than 1° and not greater than 10° .

The amount of time to remain within the tolerances must not be greater than 5 seconds.

Sensor noise must be on, $SN = 1$, but you may change the variances. Do not set any of the variances to zero. All sensors must have some noise.

Do not have more than one fuel tank with less than 1 kg of fuel. Fuel burn is must be turned on, $FB = 1$.

Submit via email by 11:59 PM on Friday April 12:

One Simulink file containing you simulation (modified from the provided plant model and named correctly)

One M-file which will initialize any variables used by your controller named:

LastName_FirstName_EE460_H8.m

One M-file containing information for a test level named:

LastName_FirstName_LVL.m

The name of the level you create should be LVL followed by your initials.

Grading Rubric: 50 points total

Complete transmission at each of 4 points: 10 pts/each if completed, 5 pts if established communication but did not complete transmission. If submitted files do not run due to errors, 0/40 points will be awarded for this part of the assignment.

Block Diagram follows rules and paperwork submitted correctly: 10 pts

If your submitted work does not incorporate the options you were assigned, 5 pts will be deducted for each option not used.

Final Project (Modify Homework 8 to refine and improve performance.)

Submit via email by 11:59 PM on Wednesday April 17:

One Simulink file containing your simulation (modified from the provided plant model and named correctly)

One M-file which will initialize any variables used by your controller named LastName_FirstName_EE460_Final.m

One report document (electronic copy required).

Presentation slides (electronic copy required).

Grading Rubric: 100 points total

Mission Success: 70 points

Each project will be run through a series of Levels and the score noted. The first 4 levels will be the same ones used for the homework assignments. Three points will be awarded for each COMM point with a completed transmission on these levels for 48 possible points. If points were lost on homework 5 through 8 because complete transmission at each of 4 points was not achieved, half of the credit for those lost points will be awarded if your final submission successfully completes those levels. The remaining 22 points for this part will be based on how well the controller performs at the more difficult levels, factoring in the relative performance of other controllers and degree of difficulty due to design options. If your submitted work does not incorporate the options you were assigned, 5 pts will be deducted for each option not used.

Presentation: 5 points

Each student will give a no more than 10 minute presentation to the class, summarizing their design and discussing what was easy or difficult especially factoring in the various design options.

Follow Simulink Rules: 10 points

Points will be deducted if the submitted Simulink diagrams do not follow the Rules posted on the course website.

Documentation: 10 points

Each student will submit a report document summarizing your design and what you learned doing this project. This document must include a one page User's Guide which will enable others to run your simulation.

Evaluation of other Projects: 5 points

During the final exam period, each student will be assigned several designs to evaluate. The student will run the simulation 5 times, then fill out an evaluation form for the controller.

Failure to show up for the evaluation (Final Exam) will result in loss of all Mission Success points